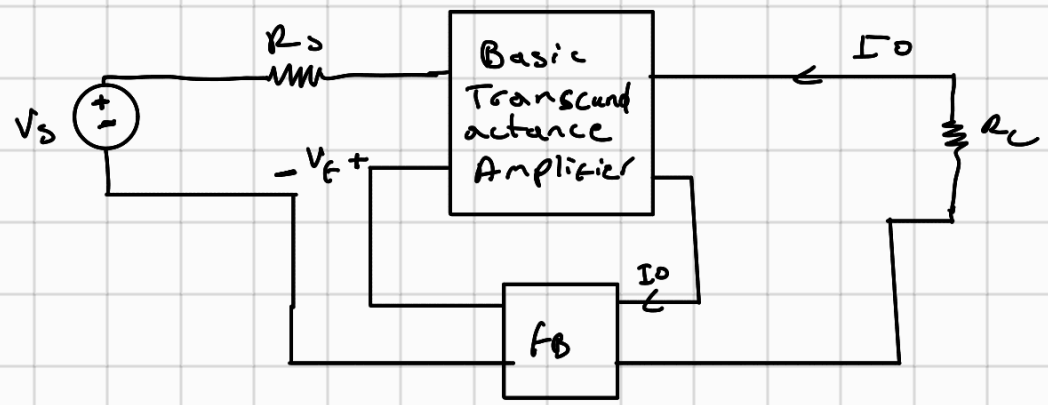
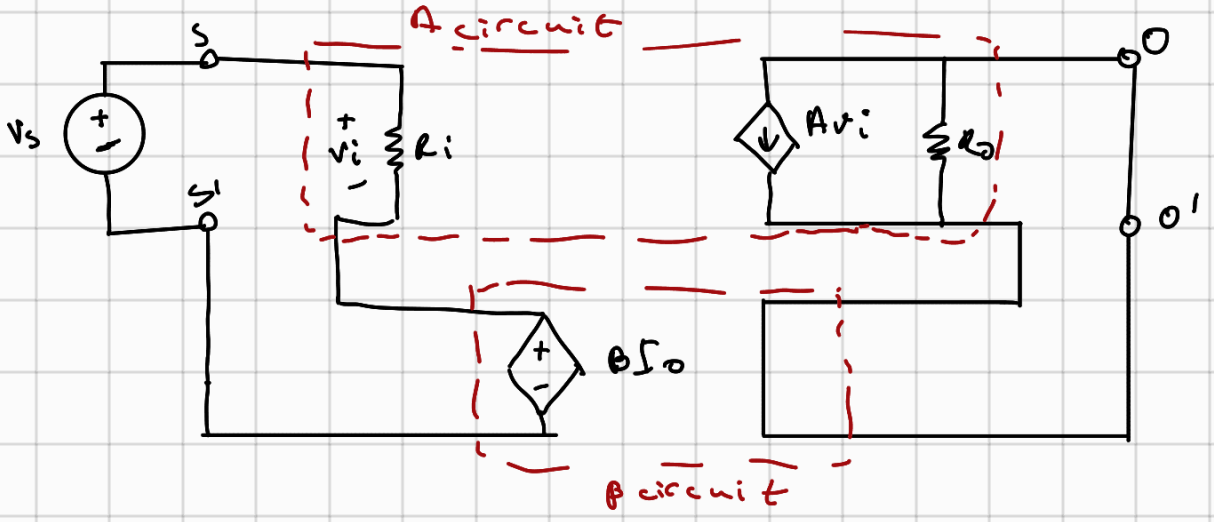
(O) (I)

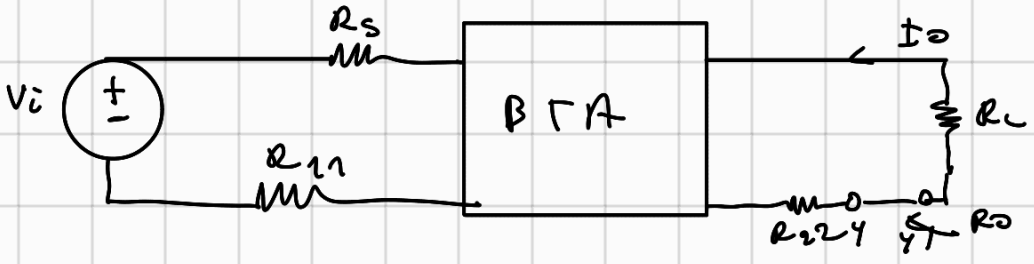
Series - Series Feedback Topology

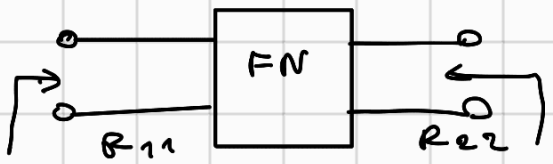
- Suitable for transconductance Amplifier



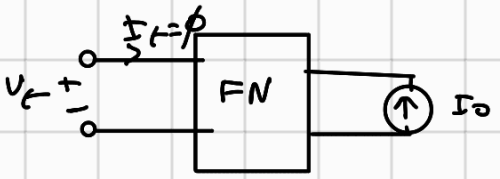


Two-port equivalent circuit





$$\frac{I_o}{V_1} = A$$



$$\beta = \frac{V_f}{I_o}$$

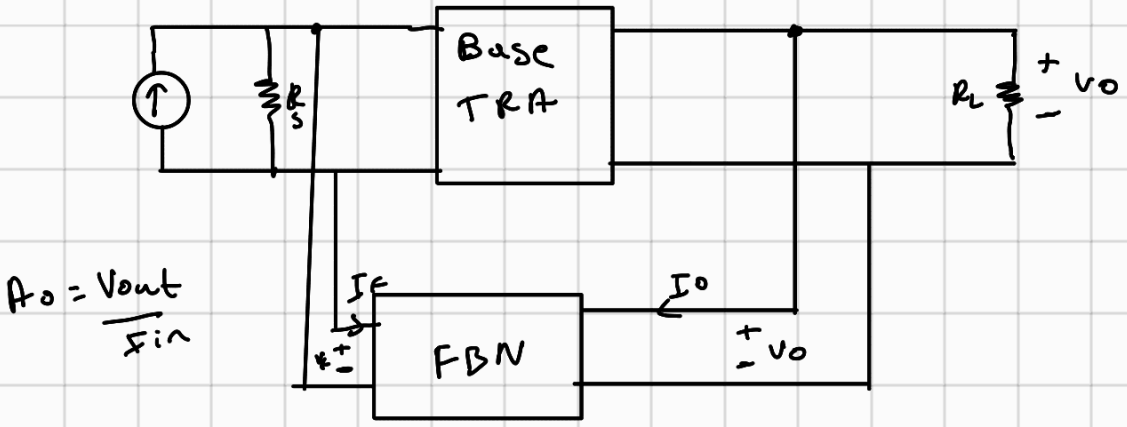
$$R_{if} = R_i(1+A\beta)$$

$$R_{of} = R_o(1+A\beta)$$

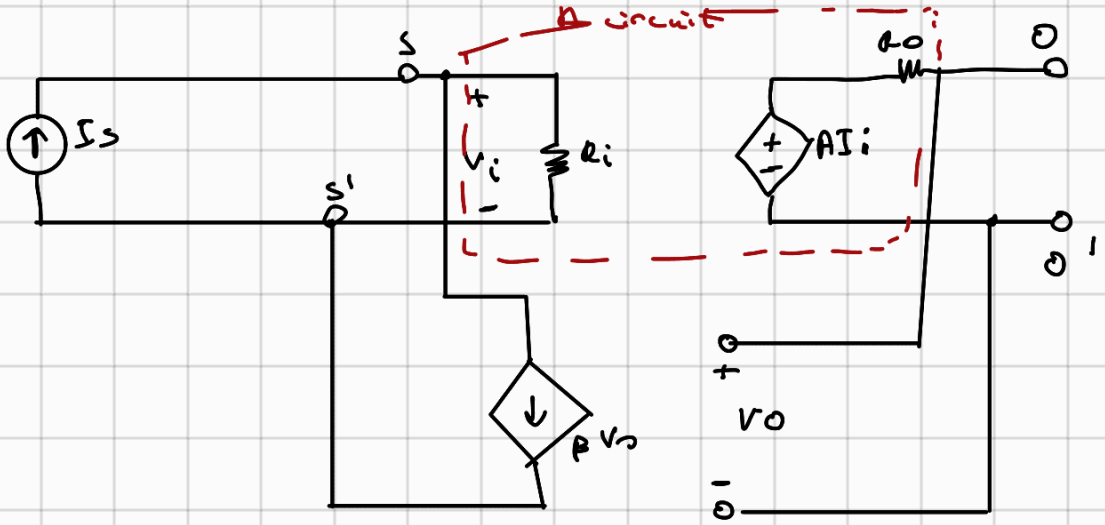
$$A_f = \frac{A}{1+A\beta}$$

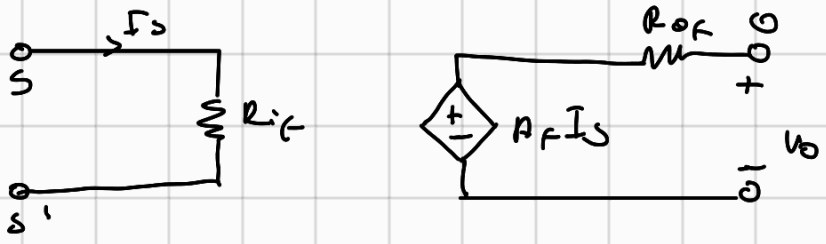
Shunt - Shunt feedback topology

(Transcresistance amplifier)

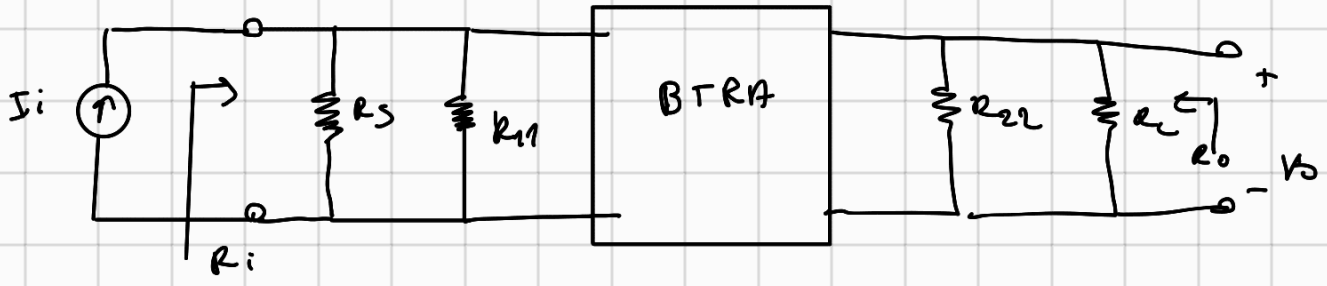


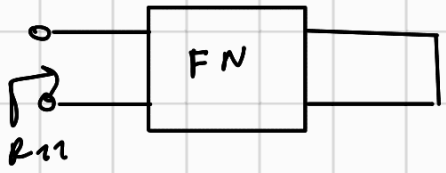
$A_o = \frac{V_{out}}{I_{in}}$

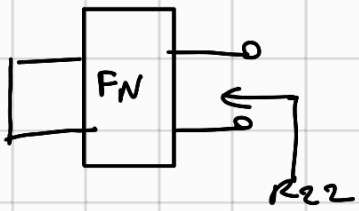


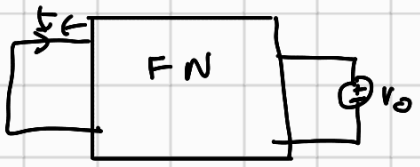


Close loop equivalent circuitmodel









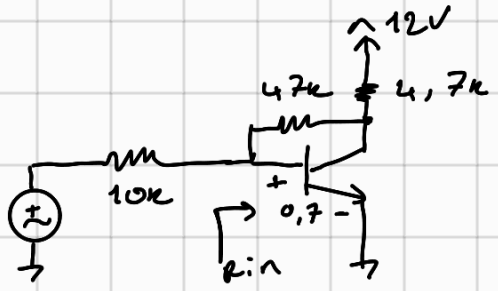
$$\beta = \frac{I_f}{V_o}$$

$$R_{if} = \frac{R_1}{1+A\beta}$$

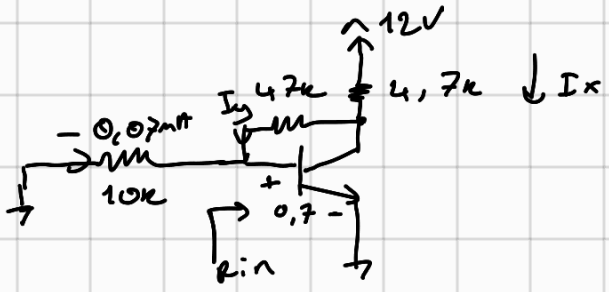
$$R_{of} = \frac{R_o}{1+A\beta}$$

$$A_f = \frac{A}{1+A\beta}$$

② Analyze the circuit below and calculate $\frac{V_o}{V_s}$, R_{in}, and R_{out}



ignore r_o | $\beta = 100$
($V_A = \infty$)



$$I_x = \frac{12 - V_c}{4{,}7k\Omega}$$

$$I_B = I_y - 0{,}07mA$$

$$I_C = 100 I_B \Rightarrow 100 I_y - 7mA = I_C$$

$$I_y = \frac{V_c - 0{,}7}{4{,}7k}$$

$$I_C = I_x - I_y = \frac{12 - V_c}{4{,}7k} - \left(\frac{V_c - 0{,}7}{4{,}7k}\right)$$

$$100\left(\frac{V_c - 0{,}7}{(4{,}7k)}\right) - 7mA = \frac{12 - V_c}{4{,}7k} - \left(\frac{V_c - 0{,}7}{4{,}7k}\right)$$

$V_C = 4,95V$

$I_D = 0,015mA$

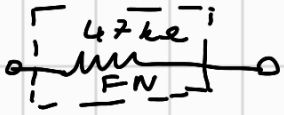
$I_C = 1,5mA$

$g_m = \frac{I_C}{V_t} = 60\,mS$

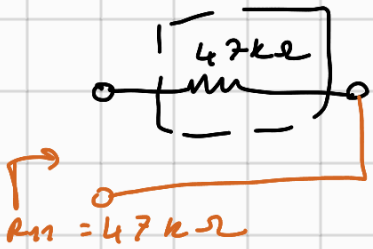
$r_\pi = (\beta+1)\cdot\frac{1}{g_m} = 1,66k\Omega$

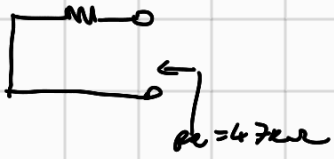
a) Feedback type: Shunt (O) – Shunt (I) (Transresistance)

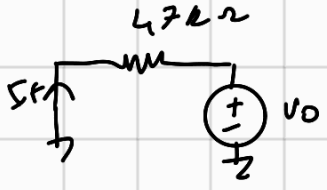
b) Feedback Network:



c) Calculate R_{11}, R_{22}, and β

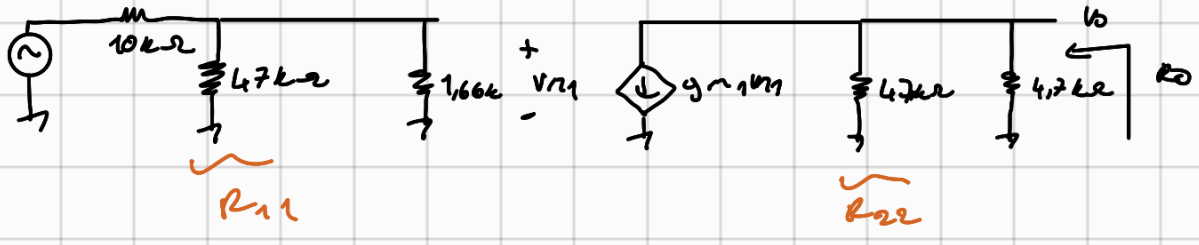


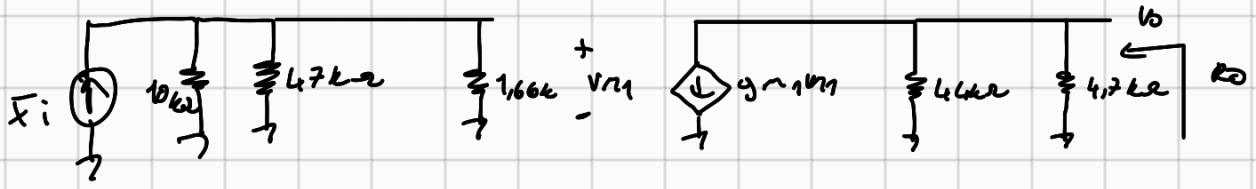




$\frac{I_f}{V_O} = \frac{-1}{4,7k\Omega} = \beta$

d)





$A = \frac{V_O}{I_i} \Rightarrow$ | $V_O = -g_{m1}\,V_{\pi 1}\,(4,27k)$ (60 mS)

$V_O = -256,2\,V_{\pi 1}$

$V_{\pi 1} = I_i\,(10k \,//\, 4,7k \,//\, 1,66k)$

$V_{\pi 1} = I_i \cdot 1,38k$

$$V_o = -256{,}2 \cdot I_i \cdot 1{,}38k$$

$$A = \frac{V_o}{I_i} = -353{,}6\ k\Omega$$

$R_i = 1{,}38k$

$R_o = 4{,}27k$

$$A_f = \frac{A}{1 + A\beta} = \frac{-353{,}6k}{1 + 353{,}6k \cdot \frac{1}{4{,}7k}}$$

$$= \frac{-353{,}6k\Omega}{5{,}52}$$

$A_f = -41{,}4\ k\Omega$

$$R_{if} = \frac{1{,}38k}{5{,}52} \Rightarrow R_{if} = 162\Omega$$

$$R_{of} = \frac{4{,}27}{5{,}52} = 501{,}2\Omega$$

$$R_{in} = \left(\frac{1}{162} - \frac{1}{10k}\right)^{-1} \approx 165\Omega$$

$R_{out} = R_{of}$

$$\frac{V_o}{I_i} = -353{,}6k\Omega //$$

$$\frac{V_o}{V_s} = \frac{A_f}{R_s} = -4{,}14\left[V/V\right]$$

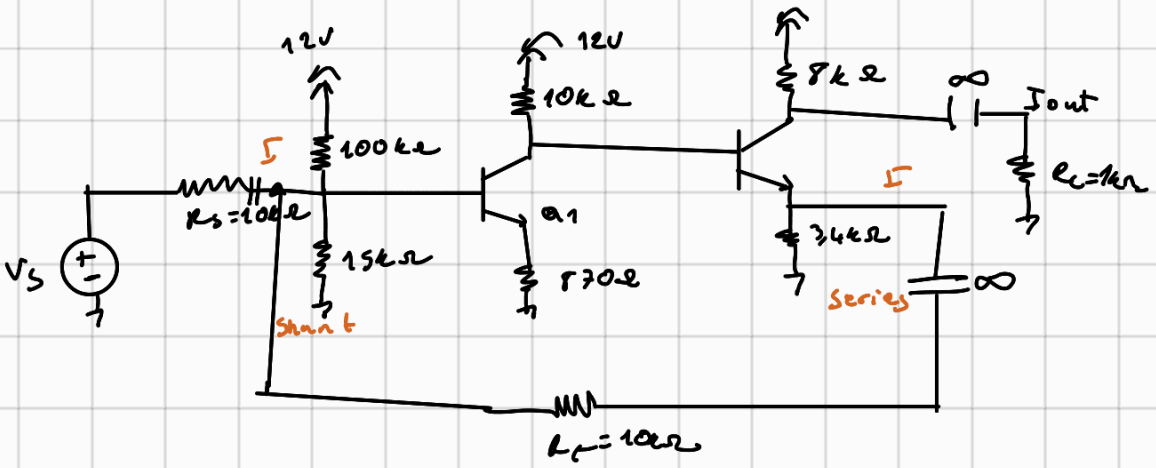
Analyze the current amplifier given below and calculate:

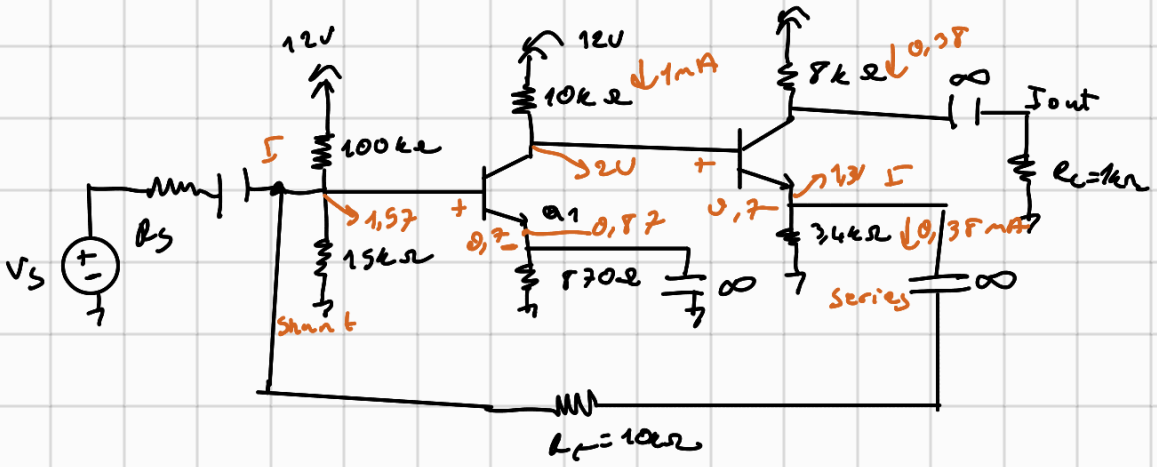
$\frac{V_o}{V_{in}}$, $\frac{I_{out}}{I_{in}}$, R_{in} and R_{out}

Assume $\beta = 100$ for small signal

∞ for DC

$V_A = 75\,V$





$I_{C1} = 1\,mA$ $\quad I_{C2} = 0{,}35\,mA$

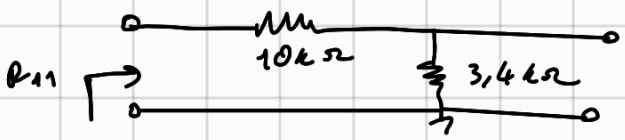
$g_{m1} = 40\,mS$ $\quad g_{m2} = 15{,}2\,mS$ $\quad r_o = \frac{V_A}{I_C}$

$r_{\pi 1} = 2{,}5\,k\Omega$ $\quad r_{\pi 2} \simeq 66\,k\Omega$

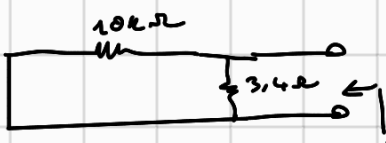
$r_{o1} = 75\,k\Omega$ $\quad r_{o2} = \frac{75}{0{,}35} \simeq 197\,k\Omega$

a) Feeback type : Series-Shunt

b) Feedback network out two port parameters

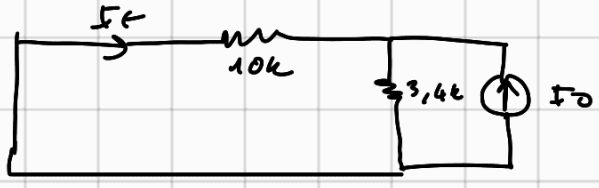


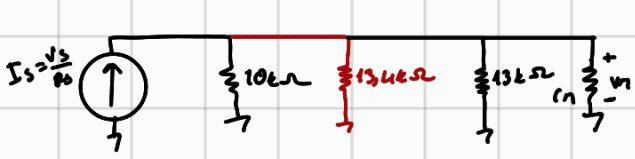
$R_{11} = 13{,}4\,k\Omega$

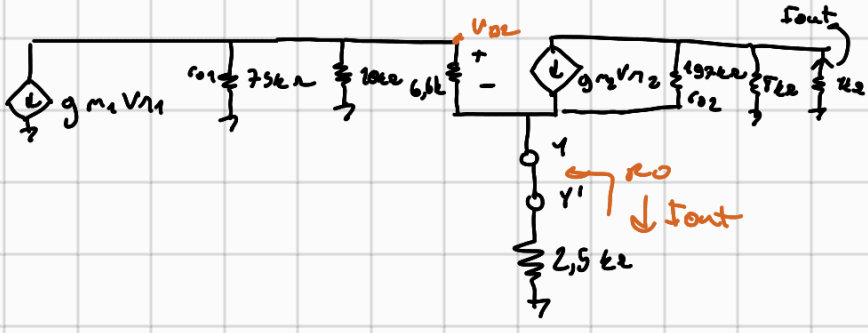


$R_{22} = 10k // 34k\Omega \quad \Rightarrow R_{22} = 25k\Omega$

$$\beta = \frac{I_f}{I_o} = \frac{-3,4k}{13,4k} \cong -0,25 \left[A/A\right]$$







$$A = \frac{I_{out}}{I_s}$$

$$V_{\pi 1} = I_s \left(10k \,//\, 13,4k \,//\, 13k \,//\, 2,5k\right) = 1,55k I_s$$

$$V_{o2} = -g_{m1} v_{\pi 1} \left(75k \,//\, 10k \,//\, \left(6,6k + 250\right)\right) \cong -320\, V_{\pi 1}$$

$(\beta+1)(2,5k)$

$$I_{out} \cong \frac{v_{be}}{\frac{6,6k\Omega}{100} + 2,5k\Omega} \cdot \frac{7}{9}$$

$2,5k\Omega \,//$

$$\frac{I_{out}}{I_s} = -192,4 \left[A/A\right]$$

$$R_i = 1,55k\Omega$$

$$R_o = 2,5k\Omega + \underbrace{\frac{6,6k\Omega}{\beta+1}}_{\sim 10\Omega} + \underbrace{10k \,//\, 75k}_{\sim 10\Omega}$$

$$\cong 2,7k\Omega$$

$$A_f = \frac{A}{1 + A\beta}$$

$$1 + A\beta = 1 + (-0,25 \times -192,4)$$

$$\boxed{1 + A\beta = 49,1}$$

$$A_f = \frac{-192,4}{49,1} \cong -3,92 \quad \left[A/A\right]$$

$$R_{iF} = \frac{R_i}{1 + A\beta} = \frac{1{,}55\,k}{49{,}1} \simeq 31{,}5\,\Omega$$

$$R_{oF} = R_o (1 + A\beta) = 2{,}7\,k \times 49{,}1 \simeq 132\,k\Omega$$